Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Maximilian Rieger	Email	maximilian.rieger@igcv.fraunhofer.de
Institute Name	Fraunhofer IGCV			
Address	Am Technologiezentrum 2, 86159 Augsburg			

2. Technology Overview

Title of Proposed Technology	Advanced thermal bending technology for fiber-reinforced thermoplastic structures			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (X), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery-materials		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / railway vehicles	Body & lightweight technology	Fiber material	Fiber-reinforced composite material
Concept of Proposed Technology	• *Development of a **thermal bending process** for continuous fiber-reinforced thermoplastic composite profiles* • *Forming of **straight thermoplastic composite profiles into complex geometries*** • *Controlled **local heating and cooling** to enable deformation without fiber damage* • *Processing of **solid and hollow profiles** with round and rectangular cross-sections* • *Adaptation of bending parameters to **material system, geometry, and bending radius*** • *Reduction of **warpage, surface defects, and geometric deviations** in bent areas* • *Design of **application-specific bending zones** optimized for load paths and functional requirements* • *Compatibility with **pultruded, filament-wound, or extruded thermoplastic composite***			

	profiles • *Integration into* ***continuous or semi-continuous production process chains*** • *Applicable to* ***automotive, rail, construction, energy, aviation and industrial lightweight structures***
Rationale for Joint R&D Project	• *Thermal bending of continuous fiber-reinforced thermoplastic composites is a* ***key enabling technology*** *for complex, load-optimized structures* • *Current bending solutions are* ***limited in industrial robustness, reproducibility, and scalability*** • *Successful implementation requires* ***combined expertise in materials, process physics, and industrial manufacturing*** • *Research institutes provide know-how in* ***process development, material behavior, and bending mechanics*** • *Korean SME involvement is essential for* ***industrialization, automation, and production-oriented optimization*** • *Joint R&D reduces* ***technical and economic risks*** *associated with scaling bending processes* • *Collaboration enables* ***adaptation of bending technology to industrial equipment, standards, and production environments*** • *Korean SMEs contribute* ***application knowledge, market access, and commercialization pathways*** • *International cooperation accelerates* ***technology maturity and time-to-market*** • *The joint project supports Korea's strategy in* ***advanced manufacturing and lightweight structures***
Key Words	1. Bending of fiber reinforced thermoplastic composites 2. Fiber-reinforced composites 3. Lightweight design 4. Cost efficient manufacturing 5. Future mobility

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	• *Develop a robust and repeatable thermal bending process for continuous fiber-reinforced thermoplastic composite profiles* • *Enable bending of solid and hollow profiles with round and rectangular cross-sections* • *Achieve controlled deformation without fiber damage or loss of mechanical performance* • *Minimize warpage, spring-back, and geometric deviations in bent areas* • *Improve surface quality and dimensional accuracy after bending* • *Define material- and geometry-specific bending parameter windows* • *Enable application-specific bending zone design aligned with load paths* • *Establish cost-efficient, production-oriented bending cycle times suitable for industrial manufacturing* • *Validate bending performance for structural and functional composite components* • *Prepare the bending technology for pilot-scale validation and industrial implementation at a Korean SME partner*			
R&D Target Specifications	***Geometric Accuracy and Quality*** • *Minimum achievable bending radius: ≤ 5 × profile height/diameter (material- and geometry-dependent)* • *Dimensional deviation after bending: ≤ ± 2 %* • *Spring-back after bending: ≤ 5 %* • *Surface damage in bending area: No visible cracks, fiber breakage, or delamination* ***Mechanical Performance Retention*** • *Retention of tensile and flexural strength after bending: ≥ 60 % of straight profile properties* ***Process Stability and Reproducibility*** • *Repeatability of bending geometry: ≥ 95 % within defined tolerances* • *Stable bending over continuous production runs without process interruptions* ***Productivity and Cycle Time*** • *Bending cycle time: ≤ 60 seconds per bending operation* • *Suitability for automated or semi-automated production* • *Low scrap rate: ≤ 5 % during steady-state operation* ***Industrial Applicability*** • *Compatibility with solid and hollow profiles (round and rectangular cross-sections)* • *Applicability to pultruded, filament-wound, or extruded thermoplastic composite profiles* • *Validation under industrially relevant production conditions*			

R&D Plan	***Research Approach*** • *Systematic development of a production-oriented thermal bending process for continuous fiber-reinforced thermoplastic composites* • *Combination of material behavior analysis, process development, and pilot-scale validation* • *Iterative optimization through experimental trials and feedback from industrial requirements* • *Close collaboration between research institute and Korean SME to align technology development with industrial implementation* ***Key R&D Tasks and Methodologies*** *Task 1: Material and Profile Preparation for Bending* • *Selection and preparation of thermoplastic composite profiles suitable for bending* • *Evaluation of solid and hollow profiles with round and rectangular cross-sections* • *Analysis of material behavior under thermal and mechanical loading during bending* *Methodologies:* • *Material screening and comparative trials* • *Thermal and mechanical characterization* • *Microscopy and defect analysis* *Task 2: Heating and Cooling Concept Optimization* • *Optimization of controlled local heating concepts for defined bending zones* • *Design of efficient cooling strategies to stabilize geometry after bending* • *Optimization of temperature gradients to avoid fiber damage and surface defects* *Methodologies:* • *Experimental heating and cooling trials* • *Temperature monitoring and control studies* • *Parameter variation and sensitivity analysis* *Task 3: Bending Process and Tooling Development* • *Design and optimization of bending tools and fixtures* • *Adaptation of bending kinematics to profile geometry and material system* • *Reduction of warpage, spring-back, and geometric deviations* *Methodologies:* • *Experimental bending trials on dedicated bending equipment* • *Tooling iteration and process tuning* • *Geometric measurement and dimensional analysis* *Task 4: Process Stabilization and Cycle Time Optimization* • *Establishment of stable bending parameter windows* • *Optimization of cycle times for cost-efficient industrial production* • *Reduction of scrap rates and process interruptions* *Methodologies:*

	• *Long-duration bending trials* • *Statistical evaluation of process stability and repeatability* • *Production-oriented cycle time analysis* *Task 5: Performance Validation* • *Evaluation of mechanical property retention after bending* • *Assessment of surface quality and structural integrity in bending zones* • *Verification of reproducibility and quality consistency* *Methodologies:* • *Mechanical testing before and after bending* • *Microscopy and surface inspections* • *Data-driven correlation of process parameters and performance* *Task 6: Pilot-Scale Demonstration and Technology Transfer Preparation* • *Pilot-scale production of representative bent composite components* • *Validation under industrially relevant production conditions* • *Preparation of process guidelines and transfer documentation for Korean SME* *Methodologies:* • *Pilot-scale bending trials* • *Documentation of best practices and parameter sets* • *Joint technical workshops and training sessions* ***Expected Milestones and Timeline*** • *Year 1: Material selection, heating and cooling concept optimization, initial bending feasibility for target geometries* • *Year 2: Tooling optimization, process stabilization, achievement of target geometric and mechanical specifications* • *Year 3: Cycle time optimization, pilot-scale validation, preparation for industrial implementation at Korean SME partner*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IGCV leverages its expertise in composite manufacturing and process optimization. With a dedicated bending machine specifically designed for continuous fiber-reinforced thermoplastic composites, Fraunhofer IGCV provides the technological foundation for advancing bending technology as a scalable and industrially viable process. In previous studies [1-3], we investigated the thermoplastic pultrusion and subsequent bending processes to gain insights into processing, providing a foundation for the proposed research topic. Key contributions: • Process development & optimization: Fraunhofer IGCV will lead the development and refinement of the bending process, ensuring high process stability, repeatability, and material integrity to meet industrial requirements. • Technology validation & scaling: Through extensive testing and iterative improvements, Fraunhofer IGCV will advance the bending process to a higher Technology Readiness Level, preparing it for industrial implementation. • Process-specific tooling & equipment development: Design and optimization of bending-specific tools, fixtures, and auxiliary systems to enable seamless integration into industrial manufacturing environments. • Material characterization & performance testing: Fraunhofer IGCV will conduct mechanical testing, microscopy, and process analytics to assess key properties such as surface quality, void content, and mechanical performance, ensuring suitability for demanding applications. • Training & knowledge transfer: Fraunhofer IGCV will provide specialized training sessions and workshops to the Korean partner, ensuring that expertise in bending technology is effectively transferred and implemented. • Technology transfer & industrialization support: Supporting the set-up and adaptation of the technology at the Korean partner's facility, ensuring a smooth transition from research to industrial application. Through these contributions, Fraunhofer IGCV will play a crucial role in developing, validating, and transferring bending technology, driving industrial innovation, and strengthening the competitiveness of composite manufacturing in global markets.
Expected R&R of Korean SME Partner	The Korean SME(s) could play a key role in industrializing and commercializing the developed bending technology for fiber-reinforced thermoplastic composites. Their primary focus could be on adapting the technology for large-scale production, developing market strategies, and ensuring the long-term economic viability of the process. Additionally, the SME(s) may explore complementary process developments and digital tools to enhance the technology's integration into industrial applications. Potential contributions and areas of focus: • Industrialization & scaling: The Korean SME(s) could support the adaptation and integration of the bending technology into an industrial production environment, ensuring process optimization for efficiency, cost-effectiveness, and quality standards. • Development of a virtual process chain & simulation tools: To enable a smooth transition to industrial production, the SME(s) could contribute to the development of a virtual process chain, including simulation and modeling of the bending process. This would provide predictive analysis, process optimization, and a deeper understanding of material behavior, reducing

	development risks and accelerating implementation. • **Commercialization & market strategy:** A key responsibility could be developing a market entry strategy, identifying target industries and applications where the bending technology can provide a competitive advantage. Potential sectors include lightweight components for automotive, aerospace, hydrogen infrastructure, and construction. • **Technology exploitation & IP utilization:** The SME(s) could lead efforts to explore patent protection, technology licensing, and potential partnerships to ensure successful commercialization of the developed process. • **Process chain optimization & potential follow-up processes:** Beyond bending, the SME(s) could investigate additional downstream processes, such as injection molding, over-molding, or automated finishing steps, to create a fully integrated production chain and expand the range of potential applications. • **Production readiness & supply chain development:** Ensuring a reliable supply chain for materials, tooling, and equipment will be critical for commercialization. The SME(s) could assist in establishing partnerships with suppliers and manufacturing service providers. • **Demonstrator development & validation:** The SME(s) could contribute to the production, testing, and validation of demonstrator components, ensuring that the technology meets industry-specific mechanical, durability, and regulatory requirements. • **Strategic networking & industry engagement:** To drive adoption of the bending technology, the SME(s) could actively engage with potential customers, suppliers, and regulatory authorities through trade fairs, industry consortia, and certification bodies. Through these potential contributions, the Korean SME(s) could help bridge the gap between research and industrialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Outcomes • Establishment of a validated thermal bending process for continuous fiber-reinforced thermoplastic composite profiles • Reliable forming of complex geometries with high dimensional accuracy and low warpage • Preservation of mechanical performance and fiber integrity after bending • Demonstration of stable, repeatable, and production-oriented bending cycles • Availability of a scalable bending technology suitable for industrial implementation Commercialization and Industrial Impact • Readiness for technology transfer and deployment at a Korean SME partner • Enablement of new lightweight composite products with complex geometries • Expansion of Korean SME capabilities in advanced thermoplastic composite manufacturing • Improved competitiveness of Korean SMEs in automotive, rail, construction, and industrial markets Economic Impact • Reduction of manufacturing costs through efficient, automated, and low-scrap bending processes • Shorter production chains by replacing multi-part assemblies with bent single-piece components • Creation of new business opportunities and value-added manufacturing for Korean SMEs Broader Impact • Support of sustainable manufacturing through efficient use of recyclable thermoplastic composites • Contribution to national strategies for advanced manufacturing and lightweight structures • Strengthening of international collaboration between Korean industry and applied research institutes • Positioning of Korean SMEs as technology leaders in thermal bending of composite structures

Impressions of »Bending Technology« at Fraunhofer IGCV

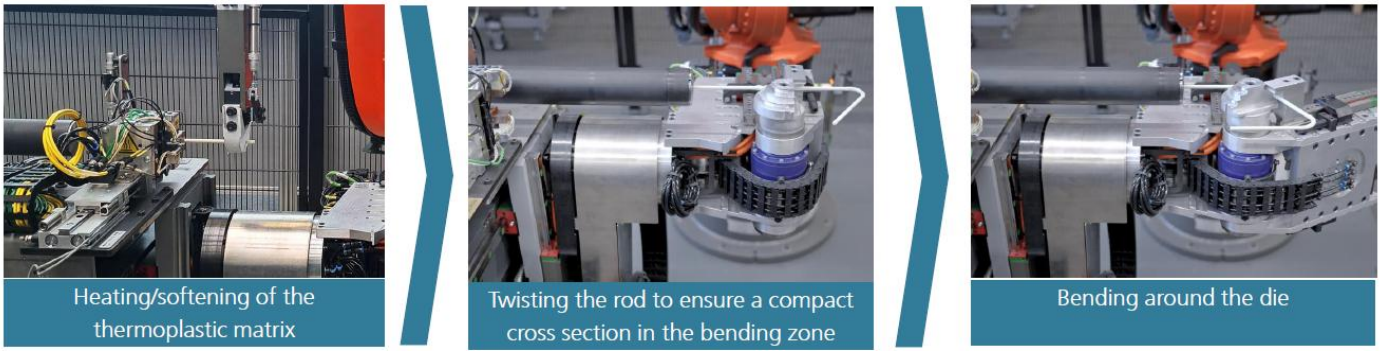

Figure 1: Process steps of »Bending Technology«

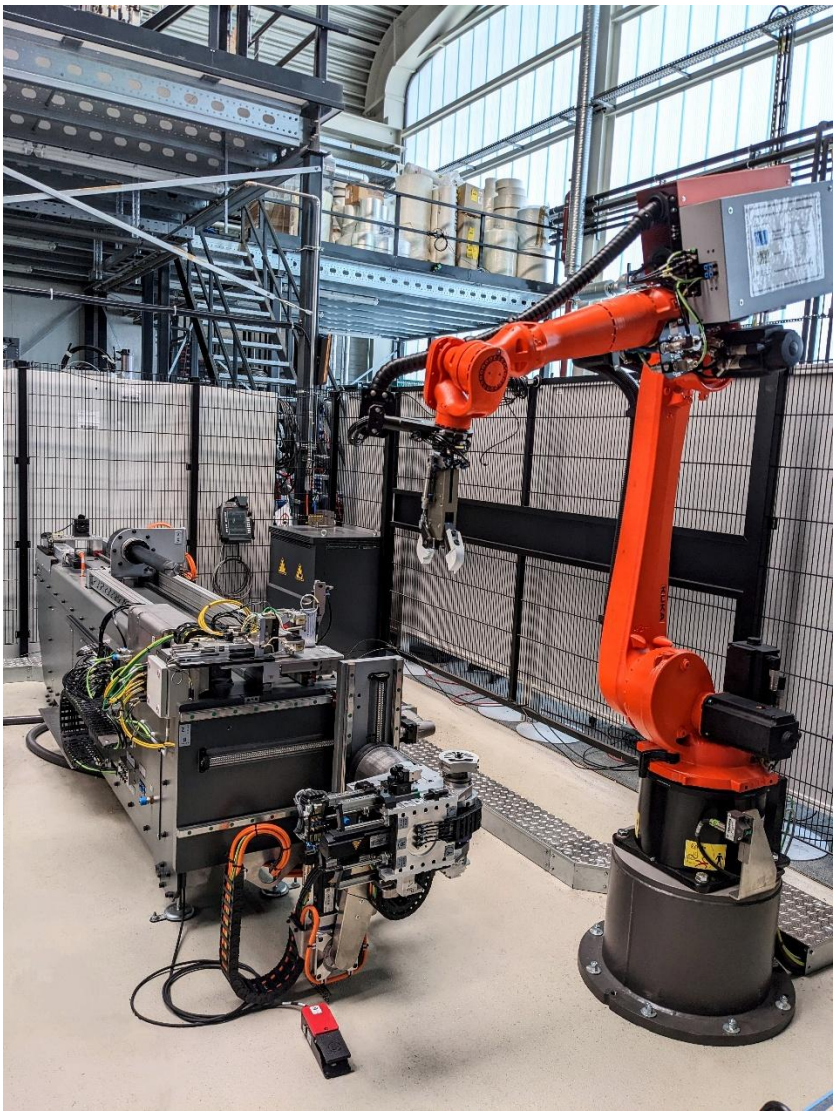

Figure 2: Bending machine »HotBend 35« at Fraunhofer IGCV

6. References

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